

This certificate is awarded to

GADIPE VISHNUVARDHAN

for successfully completing

CCNAv7: Introduction to Networks

through the Cisco Networking Academy program.



Lynn Bloomer
Director
Cisco Networking Academy

15 Apr 2024
Completion Date



Mar 13, 2022

Gadipe Vishnu Vardhan

has successfully completed

Programming Fundamentals

an online course authorized by Duke University and offered through Coursera

A handwritten signature in black ink, appearing to read "Andrew D. Hilton".

Andrew D. Hilton | Genevieve M. Lipp | Anne Bracy
Assistant Professor of the Practice | Adjunct Professor | Senior
Lecturer
Electrical and Computer Engineering | Electrical and
Computer Engineering | Computer Science, Cornell University

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This certificate is awarded to

GADIPE VISHNUVARDHAN

for successfully completing

Cybersecurity Essentials

through the Cisco Networking Academy program.



Lynn Bloomer
Director
Cisco Networking Academy

25 Dec 2023
Completion Date



dataminds
TRAINING | PLACEMENTS | CONSULTING

Specialization Certificate

has been presented to

G VISHNU VARDHAN

And has successfully completed course of study

Python Programming	Business Statistics for Data Science	Data Visualization	
Machine Learning	Deep Learning	Natural Language Processing	Computer Vision
SQL	Microsoft Power BI		Master Programme Capstone Project

hereby certified as

AI & Data Science Specialist

Director
Data Minds Analytics Pvt Ltd

Data Minds Analytics



Digital Skills



Date : 15/08/25 TO 30/11/25

Certificate ID : DSA132-RR1537



GenAI Certified Workshop By Preplnsta

Be the 1% who builds, not waits.

Right Skills. Right Strategy. Right Time.



12 June 2026
Friday



06:00
PM IST



By - Mohan KTK

Senior Trainer

Workshop Agenda — 9 In-Depth Sessions

01

Foundations of AI & Gen AI

AI family tree, history, why now?

02

How Gen AI Actually Works

Neural nets, transformers, tokens, training, RLHF

03

The Gen AI Ecosystem & Tools

Text, image, code, audio, video, agents — live demos

04

Prompt Engineering (Deep Dive)

CLEAR, CoT, ToT, patterns, system prompts, parameters

05

Hands-On Activity Sessions

Three structured challenges — beginner to advanced

06

Gen AI Across Industries

Software, healthcare, finance, education, creative, legal

07

Responsible AI & Ethics

Bias, privacy, copyright, hallucinations, regulation

08

Careers, Skills & Certifications

Roles, salaries, skills, portfolio, certifications

09

Projects & Getting Started

Beginner to advanced project ideas, resources, roadmap

SECTION 1

Foundations of AI & Gen AI

From what intelligence means to why Gen AI changed everything

What is Artificial Intelligence?

Intelligence is the ability to learn, reason, and solve problems. Artificial Intelligence is when machines do this.

Narrow AI (ANI)

AI designed for ONE specific task. Cannot go beyond its programming.

- Spam filters
- Voice assistants (Siri)
- Netflix recommendations
- Face unlock on your phone

General AI (AGI)

AI that can do ANY intellectual task a human can. Does NOT exist yet.

- Human-level reasoning
- Transfer learning across all domains
- True understanding & creativity
- Still theoretical — no system achieved this

Super AI (ASI)

AI that surpasses ALL human intelligence across every domain.

- Hypothetical future state
- Debates: beneficial vs. existential risk
- Timeline unknown
- Subject of intense research & debate

Artificial Intelligence

Machines that simulate human intelligence

Rule-based systems, expert systems, search algorithms

Machine Learning

Systems that learn from data without explicit programming

Decision trees, random forest, SVM, regression

Deep Learning

Multi-layer neural networks that learn complex patterns

Image recognition, speech-to-text, AlphaGo

Generative AI

Deep learning models that create new content — text, images, code, audio, video

ChatGPT, Gemini, DALL-E, Mid Journey, Sora

History of AI — From Dream to Reality



The Perfect Storm: Why Gen AI Exploded Now

Three forces converged at the same time — and AI went from research labs to your phone in 5 years.



Compute Power

- GPU costs dropped 1000× in 10 years
- NVIDIA built chips specifically for AI
- Cloud computing: anyone can rent AI supercomputers
- Training GPT-4 used ~25,000 A100 GPUs for months



Massive Data

- The internet created an ocean of text data
- Common Crawl: 50+ billion web pages archived
- Wikipedia, books, code repositories, Reddit, YouTube transcripts
- More data = smarter models (scaling law)



Algorithm Breakthroughs

- 2017: Transformer architecture — self-attention mechanism
- RLHF (2022): Humans teach AI to be helpful
- Scaling laws: bigger models + more data = better performance
- Open-source: Llama, Mistral made AI accessible to all

Types of Generative AI Models

Large Language Models

(LLMs)

ChatGPT, Claude, Gemini, Llama

Generate, summarize, translate, reason with text. Trained on trillion+ words.

Diffusion Models

(Image Gen)

DALL-E, Midjourney, Stable Diffusion

Create images from text prompts by gradually denoising random pixels.

Code Models

(Code Gen)

GitHub Copilot, Cursor, CodeLlama

Write, debug, explain, and review code in 100+ programming languages.

Audio Models

(Audio AI)

ElevenLabs, Suno, Whisper, MusicLM

Clone voices, generate music, transcribe speech, create sound effects.

Video Models

(Video Gen)

Sora, RunwayML, Pika, Kling

Generate and edit videos from text or images. Minutes of footage from a sentence.

Multimodal Models

(Any-to-Any)

GPT-4o, Gemini 1.5, Claude 3

See, hear, speak, and reason across text + images + audio simultaneously.

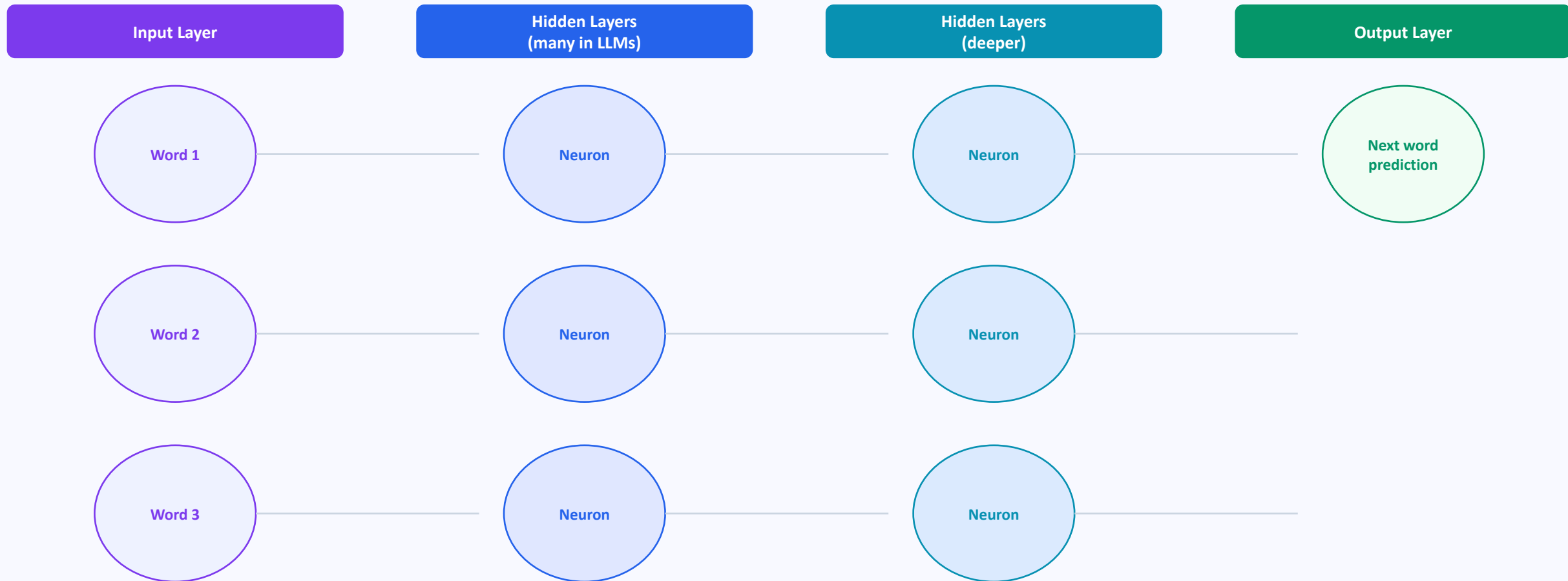
SECTION 2

How Gen AI Actually Works

From neurons to transformers — the architecture behind the magic

Neural Networks: The Building Block

A neural network is a system of interconnected nodes (like brain neurons) that learns patterns by adjusting connection strengths.



Key insight: The network adjusts millions of "weights" (connection strengths) during training until it can predict outputs correctly. Modern LLMs have BILLIONS of these weights.

The Transformer: Architecture That Changed Everything

"Attention Is All You Need" — Google Brain, 2017. This 8-page paper is the foundation of every modern LLM.

Self-Attention

The model weighs how relevant each word is to every other word in a sentence simultaneously — not one word at a time like older models.

e.g. "The bank can guarantee deposits will eventually cover future tuition costs." — Which "bank"?

Parallel Processing

Unlike older RNN models (which read left to right), transformers process ALL words at once. This makes training massively faster on GPUs.

e.g. Reading the entire sentence simultaneously vs. word by word — 100× speed improvement.

Positional Encoding

Since all words are processed at once, the model needs to know word ORDER. Special math encodes position into each token representation.

e.g. "Dog bites man" vs "Man bites dog" — same words, very different meaning.

Multi-Head Attention

Multiple attention mechanisms run in parallel, each focusing on different types of relationships (grammar, meaning, context, etc.).

e.g. One head learns syntax, another learns semantics, another learns coreference.

What is a Token?

Text is broken into chunks called tokens before an LLM processes it.

"Hello" → 1 token

"ChatGPT" → 2 tokens: "Chat" + "GPT"

"Artificial" → 3 tokens: "Art"+"ific"+"ial"

~1 English word → ≈ 1.3 tokens on average

1000 tokens → ≈ 750 English words

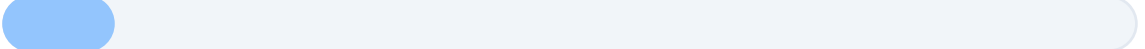
What are Parameters?

Parameters are the learned "weights" in a neural network — the model's memory of everything it learned.

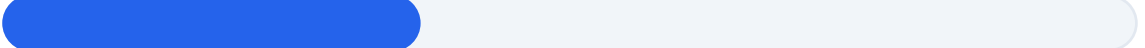
GPT-1 (2018) **117 Million**



BERT (2018) **340 Million**



GPT-3 (2020) **175 Billion**



GPT-4 (2023) **~1.8 Trillion**



Llama 3 (2024) **405 Billion**



How an LLM is Trained: Step by Step

1

Data Collection

Collect trillions of words from the internet, books, Wikipedia, code repositories, scientific papers. Filter for quality.

2

Pre-training

Feed all data through the neural network. The model learns to predict the next token. This can take months on thousands of GPUs.

3

Fine-tuning (SFT)

Train on curated examples of good conversations. Human trainers write ideal Q&A pairs. Model learns the target behaviour.

4

RLHF

Human raters rank model outputs. A reward model learns what "good" looks like. The LLM is tuned to maximize reward scores.

5

Safety Testing

Red-teaming: adversarial testing for harmful outputs, bias, and jailbreaks. Multiple rounds before public release.

6

Deployment

Model served via API or chat interface. Continues to receive user feedback for future improvement cycles.

Reinforcement Learning from Human Feedback — the technique that turned powerful but raw language models into useful assistants.



 RLHF is why ChatGPT feels "friendly" and helpful rather than just statistically predicting words. It literally learned what humans prefer.

Hallucination

AI confidently states false information — fake citations, wrong dates, invented people. It predicts plausible-sounding text, not necessarily true text.

How to handle: Always verify facts. Use Perplexity AI for research (it cites sources). Tell the AI: "Only tell me what you know for certain."

Bias in Training Data

The internet contains human biases. Models trained on this data inherit those biases — gender, racial, cultural, political — even unintentionally.

How to handle: Be aware when using AI for sensitive decisions. Use diverse prompting, specify you want balanced views, audit outputs critically.

Training Cutoff

LLMs have a knowledge cutoff date. Events after that date are unknown to the model. GPT-4 knows up to early 2024. Claude's cutoff varies.

How to handle: For recent news/events, use Perplexity AI or Gemini with Google Search enabled. Specify the date in your prompt.

Context Window Limits

Every LLM can only "see" a limited amount of text at once (its context window). Older info in a long conversation may be "forgotten."

How to handle: Break very long tasks into chunks. Use models with large context windows (Claude: 200K tokens, Gemini: 1M tokens).

SECTION 3

The Gen AI Ecosystem & Tools

From ChatGPT to AI Agents — the full landscape with live demos

Text AI: The LLM Landscape



TOOLS

C

ChatGPT

by OpenAI · GPT-4o

Most versatile. Best all-rounder for writing, coding, Q&A

💰 Free (GPT-4o-mini) / \$20 mo (Plus)

Most Popular

C

Claude

by Anthropic · Claude 3.5 Sonnet

Best for long docs, nuanced reasoning, safety-first design

💰 Free tier / Pro \$20 mo

Best Reasoning

G

Gemini

by Google · Gemini 1.5 Pro

Deep Google integration, 1M token context, web access

💰 Free tier / Advanced \$20 mo

Google Native

L

Llama 3

by Meta · Open Source

Run locally, fully open, customizable, no API costs

💰 100% Free & Open Source

Open Source

M

Mistral

by Mistral AI · Mixtral 8x7B

Lightweight, fast, great for European data privacy use

💰 Free API tier available

Efficient

P

Perplexity

by Perplexity AI · Various LLMs

Real-time web search + citations. Best for research tasks

💰 Free / Pro \$20 mo

Research

Image Generation AI

DALL-E 3 — OpenAI

Best prompt understanding, integrated in ChatGPT Plus. Natural language → images.

Midjourney — Midjourney Inc.

Highest quality artistic images. Used by designers & studios. Discord-based.

Stable Diffusion — Stability AI

100% open source. Run locally or in cloud. Infinite customization.

Adobe Firefly — Adobe

Commercial-safe (trained on licensed images). Built into Photoshop.

Code & Developer AI

GitHub Copilot — Microsoft/GitHub

Inline code completion inside VS Code. Best for auto-complete & boilerplate. \$10/month.

Cursor — Anysphere

AI-first IDE. Chat with your codebase, edit entire files, auto-debug. \$20/month.

Replit AI — Replit

Browser-based IDE with AI. Best for beginners & students. Free tier available.

Amazon CodeWhisperer — AWS

Free for individual developers. Strong on AWS services & cloud code.

Devin (AI) — Cognition AI

First "AI software engineer". Takes task → writes, tests, deploys code autonomously.

An AI Agent is an LLM that can plan, take actions, use tools, and complete multi-step tasks autonomously — not just answer questions.

Regular LLM

- ✗ You ask a question
- ✗ Gets one prompt
- ✗ Responds with text
- ✗ Cannot use tools or browse web

AI Agent

- ✓ Given a goal, it plans steps
- ✓ Breaks task into sub-tasks
- ✓ Searches web, reads files, writes code
- ✓ Executes → reviews → revises until done

What Agents Can Do:

Web Search

Perplexity, Bing

Write & Run Code

Python, JS

Read/Write Files

Docs, sheets

Send Emails

Gmail, Outlook

Browse Websites

Navigate & click

Call APIs

Any external service

Get IBM & Google Cloud Certification at Lowest Price Ever 20% Off

With
Preplnsta
Prime

Use Code

KTKMAX

And Get

20% Off

SECTION 4

Prompt Engineering Deep Dive

The most valuable skill of the Gen AI era — master it completely

Every expert-level prompt has 6 components. Master all 6 and your AI outputs will transform instantly.

01 Role / Persona

Tell the AI WHO to be. Shapes tone, depth, vocabulary & perspective.

e.g. *"Act as a senior Python developer with 10 years of experience reviewing student code."*

03 Task

The actual instruction. Use action verbs: Explain, Write, Compare, List, Debug.

e.g. *"Explain the difference between Stack and Queue with a real-world analogy."*

05 Examples (Few-shot)

Show before you tell. Single best technique to boost output quality.

e.g. *"Here is an answer I consider perfect: [paste example]. Now apply the same style to..."*

02 Context

Background the AI needs to give YOU a relevant, personalised answer.

e.g. *"I am a 3rd-year B.Tech student preparing for TCS NQT. I struggle with recursion."*

04 Format

Specify structure: bullets, table, JSON, numbered steps, code block.

e.g. *"Use bullet points. Include a comparison table. Keep it under 200 words."*

06 Constraints

Set the limits: reading level, length cap, topics to avoid, language.

e.g. *"No jargon. Suitable for a beginner. Do NOT use recursion in the explanation."*

The CLEAR Framework — Your Prompting System



FRAMEWORK

C

Context

Who are you? What's the situation? What does the AI need to know about your background, goal, or project?

"I am a final-year B.Tech CS student preparing for Infosys campus placement. I have 2 projects on GitHub but no internship experience."

L

Length & Format

Specify the exact output structure. Bullet points? Table? Code block? Word limit? Header/subheader structure?

"Format: 5 bullet points. Each point 2 sentences max. Include one code example in Python. Total under 300 words."

E

Examples (Few-shot)

Show the AI exactly what "good" looks like before asking. This is the single most impactful technique for improving output quality.

"Here is an example answer I consider perfect: [paste example]. Now apply the same style to answer: [your question]"

A

Action & Role

"Act as X and do Y." Assign a persona to shape expertise, tone, and depth. Then give a precise action verb instruction.

"Act as a senior FAANG interviewer. Evaluate my answer to this LeetCode problem and tell me exactly what I did wrong."

R

Refine Iteratively

The first response is rarely the best. Follow up to adjust tone, add detail, fix errors, try alternatives, or shorten the output.

"Good — now make it 50% shorter and more casual in tone. Replace any jargon with plain English."

0 Zero-Shot

Give the AI a task with NO examples. Relies entirely on training knowledge.

Best for: Simple, well-defined tasks where the expected format is obvious.

Prompt:

*Classify this review as Positive/Negative/Neutral:
"The product arrived late but works perfectly fine."*

Output: Neutral

Works well ~70% of the time for clear tasks.

1 One-Shot

Provide ONE example before asking. Helps the AI match format and style.

Best for: When you have a specific format or style you want matched.

Prompt:

*Example: "The laptop is amazing" → Positive
Now classify: "The delivery was slow but the product is good"*

Output: Mixed (learns from example)

Improves accuracy by ~15-25%.

3 Few-Shot

Provide 3-5 examples before the task. Dramatically improves output quality.

Best for: Complex classification, writing style matching, structured output, code generation.

Prompt:

*"It's fantastic!" → Positive
"Absolutely terrible!" → Negative
"It's okay I guess" → Neutral
Now: "Late delivery, great product" → ?*

Output: Mixed (learned nuance from 3 examples)

Best technique. Improves accuracy by 30-50%.

Make the AI reason step by step. Adding "Let's think step by step" can increase accuracy on reasoning tasks by 30-40%.

Chain-of-Thought (CoT)

Works by: Asking the model to show its reasoning before giving the final answer

- Trigger phrase: "Think step by step before answering"
- Or: "Explain your reasoning, then give the final answer"
- Best for: Math problems, logic puzzles, multi-step reasoning
- Research result: 540B parameter model + CoT > human average on MATH

Example Prompt:

*"If I have 3 apples and give 1 to my friend, then buy 4 more and eat 2, how many do I have?
Think step by step."*

Tree-of-Thought (ToT)

Extends CoT: Explores **MULTIPLE** reasoning paths simultaneously

- Like a tree: root (problem) → branches (approaches) → best leaf (answer)
- AI generates 3-5 different solution paths, evaluates each, picks the best
- Best for: Complex problems with multiple valid approaches

Example Prompt:

"Consider 3 different approaches to solve this system design problem. Evaluate pros and cons of each, then recommend the best solution."

ReAct (Reason + Act)

AI reasons about what to do, then acts (calls a tool/API), then reasons about the result. Foundation of AI agents.

Self-Consistency

Ask the same question 3 times in different ways. The answer that appears most often is likely correct.

Negative Prompting

"Do NOT use jargon. Do NOT give generic advice. Do NOT write more than 150 words." Constraints sharpen outputs.

Persona Injection

"Pretend you are a harsh but fair technical interviewer at Google. Be critical, specific, and constructive."

Prompt Transformation: Weak → Powerful

✗ WEAK

"Explain machine learning"

✓ STRONG

"Act as an ML professor teaching 2nd-year engineering students. Explain machine learning in exactly 5 bullet points. Use one real-world analogy. Avoid all mathematical notation. End with one practical application in healthcare."

✗ WEAK

"Write code for sorting"

✓ STRONG

"Act as a Python tutor. Write a Python function that sorts a list of student names alphabetically. Add detailed comments explaining each line. Show a usage example at the bottom. Target audience: first-year students learning Python."

✗ WEAK

"Help with my resume"

✓ STRONG

"Act as a senior HR manager at TCS who has reviewed 500+ resumes. I am a fresh B.Tech graduate with: 1 internship, 3 projects (Python, ML, Web dev), 8.2 CGPA. Review my resume summary and rewrite it to highlight my strongest points in 3 impactful sentences."

SECTION 5

Hands-On Activity Sessions

Three structured challenges — beginner, intermediate, advanced

Activity 1: Beginner Prompting Challenge



HANDS-ON

Tool: ChatGPT / Gemini / Claude | Work individually | Share your best result at the end

1

Role Switch Challenge

Ask an AI to explain "What is the Internet?" — first as a teacher explaining to a 5-year-old, then as a network engineer giving a technical answer, then as a comedian making it funny. Compare all three.

2

CLEAR Framework Practice

Use ALL five CLEAR components to craft a prompt asking AI to help you prepare for a GATE exam question on Operating Systems. The output must be structured, specific, and actionable.

3

Prompt Debugging

You are given this bad prompt: "Tell me about Python." Identify every element that is missing, rewrite it using the CLEAR framework targeting campus placement preparation, and share your improved output.

4

Few-Shot Learning

Create a few-shot prompt that classifies a student's answer as "Excellent / Good / Needs Improvement" — provide 3 examples then ask the AI to classify a new answer you write.

Bonus: Best prompt from each task gets shared with the class!

Activity 2: Advanced Real-World Challenge



HANDS-ON

Work in pairs | Use any AI tool | Iterate at least 3 times on your prompt

A

Build a Study Plan

Use AI to generate a 4-week personalised study plan for your weakest subject this semester. Include daily topics, resources, and a self-assessment quiz for each week. Use few-shot + CLEAR framework.

B

Career Document Creator

Create a complete job application package using AI: (1) A cover letter for a Python Developer internship (2) 3 bullet points for your resume projects section (3) Answers to 3 common interview questions. Show all prompts used.

C

Product Idea Validator

You have an app idea for engineering students. Use AI as a market researcher, business analyst, and devil's advocate to: analyze the market, find 3 competitors, identify 3 risks, and suggest 1 improvement. Chain 4 prompts together.

Activity 3: Build a Mini AI Project



HANDS-ON

Choose ONE of these no-code AI projects. Build it in 20 minutes using just AI tools and your creativity. Present what you built.



AI Study Buddy Chatbot

Beginner

Craft a detailed system prompt that turns ChatGPT into a study buddy for your department. It should: quiz you on topics, explain concepts simply, and give exam tips. Test it with 5 questions.



Content Creation Pipeline

Intermediate

Use AI to write a LinkedIn post announcing this workshop, a tweet thread (5 tweets) about what you learned today, and a short blog post intro. Show how you refined the prompts each time.



AI Problem Solver

Intermediate

Pick a real problem at your college (timetable, mess feedback, lost-and-found, etc.). Use AI to: analyze the problem, propose 3 solutions, write a proposal email to the principal, and create a simple app outline.



Prompt Cookbook

Advanced

Create a "Prompt Cookbook" — a document with 10 ready-to-use prompts for engineering students covering: study, coding, resume, interviews, projects, emails, presentations, research, and time management.

SECTION 6

Gen AI Across Industries

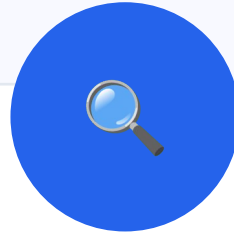
Where AI is being deployed right now — and what it means for your future career

55% of developers use AI tools daily. GitHub Copilot users are 55% faster. AI doesn't replace developers — it supercharges them.



Code Generation

Write boilerplate, functions, classes, APIs from natural language descriptions. Copilot generates 46% of GitHub code.



Code Review & Refactoring

AI reviews PRs, finds bugs, suggests better patterns, refactors legacy code, explains complex functions.



Test Generation

Automatically generate unit tests, edge cases, and test data. Reduces manual testing time by 60-70%.



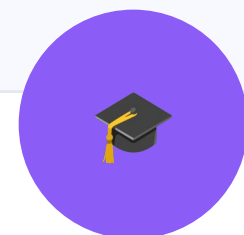
Documentation

Docstrings, README files, API documentation, changelogs — all generated automatically from code.



Debugging

Paste an error → AI explains the cause, suggests fixes, and provides a fixed version. Faster than Stack Overflow.



Learning & Upskilling

Explain unfamiliar code, new languages, frameworks. Your AI pair programmer teaches while it codes.



Healthcare

Medical Imaging

AI reads X-rays, MRIs, CT scans with accuracy exceeding radiologists for specific conditions.

Drug Discovery

DeepMind's AlphaFold solved protein folding — a 50-year unsolved problem. Revolutionizing drug discovery.

Clinical Documentation

AI listens to doctor-patient conversations and auto-generates clinical notes, saving 2-3 hours daily.

Personalized Treatment

Analyze patient history, genetics, and literature to suggest personalized treatment protocols.



Finance

Fraud Detection

Real-time analysis of millions of transactions per second to flag anomalous patterns. 90%+ accuracy.

Algorithmic Trading

AI models analyze market data, news, and sentiment to execute trades in microseconds.

Risk Assessment

LLMs read financial statements, news, and filings to assess credit risk far faster than analysts.

Customer Service

AI chatbots handle 80% of banking queries: balance checks, transfers, loan status, without human agents.

Education

► Personalised Tutoring:

AI tutors adapt to each student's pace, style, and gaps. Khan Academy's Khanmigo is already in 50+ US school districts.

► Instant Feedback:

AI grades essays, code assignments, and presentations — giving detailed feedback in seconds rather than days.

► Content Generation:

Teachers use AI to generate lesson plans, quiz questions, worksheets, and explanations in seconds.

► Language Learning:

Duolingo's AI roleplay feature lets users practice conversations with AI characters. 40% faster learning.

► Accessibility:

AI tools provide real-time captions, translations, and text-to-speech — democratising education globally.

Creative Industries

► Film & Video:

Sora generates movie-quality video. Studios use AI for de-aging actors, removing backgrounds, and VFX.

► Music Production:

Suno AI creates full songs from text. Artists use AI for composition, mixing, mastering, and sound design.

► Graphic Design:

Midjourney/DALL-E generate brand visuals. Canva AI creates presentations, social posts in seconds.

► Game Development:

AI generates textures, 3D models, NPC dialogue, and even entire game levels from text prompts.

► Writing & Journalism:

AP News uses AI for earnings reports. Writers use Claude/GPT for drafts, editing, research, interviews.

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Prime

Use Code

KTKMAX

And Get

20% Off

SECTION 8

Careers, Skills & Certifications

The complete guide to building your AI career starting today

97M

New roles needing
AI skills by 2025

40%

Jobs transformed
by AI by 2027

3×

Salary premium for
AI-skilled workers

6 LPA

Min starting salary,
AI/ML fresher (India)

Top AI Hiring Companies in India (2025)

TCS AI Cloud

Infosys Nia

Wipro AI360

Accenture AI

Google India

Microsoft India

Amazon (AWS AI)

Flipkart

Zepto

Meesho

NVIDIA India

IBM India

Top Gen AI Career Roles — Detailed Breakdown

[CAREERS](#)

AI/ML Engineer

Salary: 8–25 LPA

Builds, trains, and deploys ML models for production. Most in-demand role.

Skills: Python, TensorFlow, PyTorch, MLOps, cloud

Prompt Engineer

Salary: 6–18 LPA

Designs and optimizes prompts for LLM-powered products. Fastest growing title.

Skills: LLM knowledge, NLP, prompt frameworks, evaluation

Gen AI Developer

Salary: 10–30 LPA

Builds applications using LLM APIs — chatbots, agents, RAG systems.

Skills: Python, LangChain, APIs, RAG, vector databases

Data Scientist (AI)

Salary: 8–22 LPA

Analyses data and builds predictive models. AI tools 3× their productivity.

Skills: Python, SQL, stats, ML, visualization, storytelling

AI Product Manager

Salary: 12–35 LPA

Defines AI product strategy and roadmap. Bridges business and AI teams.

Skills: Product sense, LLM understanding, user research, metrics

MLOps Engineer

Salary: 10–28 LPA

Deploys and maintains ML models in production. Ensures reliability at scale.

Skills: Docker, Kubernetes, CI/CD, model monitoring, cloud

AI Research Scientist

Salary: 15–50 LPA

Advances state of the art in AI. Works at Google Brain, DeepMind, OpenAI, IITs.

Skills: PhD preferred, deep ML theory, math, publications

AI Ethics & Policy

Salary: 8–20 LPA

Ensures responsible AI deployment. Growing role at governments and corporations.

Skills: Policy, law, ethics, AI understanding, communication

Essential AI Skills

Technical (for engineering roles)

- Python (must-have — NumPy, Pandas, scikit-learn)
- API usage: OpenAI, Anthropic, Google APIs
- LangChain / LlamaIndex for RAG applications
- Basic ML: regression, classification, neural networks
- Git, cloud basics (AWS/GCP/Azure)

Non-Technical (for all roles)

- Advanced Prompt Engineering
- Critical evaluation of AI outputs
- AI tool proficiency (5+ tools)
- Data literacy & visualization (basics)
- Communication & AI ethics understanding

Building Your Portfolio

► GitHub Profile

Push every AI project here. Pin your 3 best. Write detailed README files using AI.

► LinkedIn Presence

Add "Prompt Engineering" and "Gen AI" to skills. Post weekly about what you're learning.

► Project Demos

Record a 2-minute demo video of each project. Host on YouTube. Embed in LinkedIn/resume.

► Certifications

Google AI Essentials, Azure AI-900, DeepLearning.AI certificates on your profile.

► Written Content

Write a blog or LinkedIn post about each project. Explains thinking AND shows communication skills.

Top AI Certifications — Complete Guide



CERTIFICATIONS

Google

AI Essentials

Beginner

Cost: FREE · Duration: 6 hours

Value: High — Google brand recognition

DeepLearning.AI

ChatGPT Prompt Engineering for Developers

Beginner

Cost: FREE · Duration: 2 hours

Value: Very High — Andrew Ng's brand

Microsoft

Azure AI Fundamentals (AI-900)

Beginner

Cost: ~₹3,500 · Duration: 40 hours prep

Value: Very High — Microsoft/Azure jobs

IBM

AI Foundations for Everyone

Beginner

Cost: Audit Free · Duration: 12 hours

Value: Medium — good for non-tech roles

Hugging Face

NLP Course with Transformers

Intermediate

Cost: FREE · Duration: 30+ hours

Value: High — industry standard tool

NVIDIA

Generative AI with LLMs (Coursera)

Intermediate

Cost: ₹3,000/mo · Duration: 20 hours

Value: High — NVIDIA brand + Coursera cert

AWS

ML Specialty Certification

Advanced

Cost: ~₹27,000 · Duration: 6-12 months prep

Value: Very High — cloud jobs globally

Google

ML Engineer Professional Cert

Advanced

Cost: ~₹27,000 · Duration: 6+ months prep

Value: Very High — top-tier engineering

Your 90-Day AI Learning Roadmap



ACTION PLAN

Days 1–30

Foundation

Complete Google AI Essentials (free, 6 hrs)

Use ChatGPT/Claude daily — 30 min/day

Practice 5 CLEAR prompts every day

Read: "The Age of Gen AI" newsletter

Join AI communities on Reddit, Discord

Complete DeepLearning.AI Prompt Engineering

Days 31–60

Build Skills

Learn Python basics (if not already — 2 hrs/day)

Build Project 1: AI-powered study quiz bot

Complete Hugging Face NLP basics (free)

Explore LangChain basics on YouTube

Set up GitHub profile — push your project

Start preparing for Azure AI-900 exam

Days 61–90

Get Credentialed

Earn Azure AI-900 or Google AI cert

Build Project 2: Chatbot with your college data

Add AI skills to LinkedIn & resume

Apply to 5 AI-related internships/jobs

Write a blog about your learning journey

Join a hackathon or open-source AI project

SECTION 9

Projects & Getting Started

From your first AI script to your first AI product

Beginner AI Projects (No Code Required)



PROJECTS

These projects require only AI tool access, a browser, and your creativity. No coding needed.

AI Content Pipeline

Use AI to create a full social media content calendar for a student club: 1 month of posts, 4 LinkedIn articles, 2 email newsletters.

Skills: Content creation, prompt chaining

Prompt Cookbook

Create a document with 20 ready-to-use prompts for engineering students — covering studying, coding, resume, interviews, emails, and projects.

Skills: Prompt engineering, documentation

AI Research Assistant

Use Perplexity AI to research a technical topic in your field. Write a 1-page summary. Document your research prompts and how you verified accuracy.

Skills: Research prompts, source verification

Personal Knowledge Bot

Create detailed context about yourself (skills, projects, interests, goals). Build a system prompt so ChatGPT can answer questions about you like a personal assistant.

Skills: System design, context engineering

AI-Powered Chatbot (Python)

Intermediate

Build a chatbot using OpenAI API + Python. Add conversation history, system prompt, and a simple web UI using Streamlit. Deploy on Hugging Face Spaces (free).

Stack: Python, OpenAI API, Streamlit

RAG Document Q&A System

Intermediate

Build a system that answers questions from your own documents (lecture notes, textbooks). Uses LangChain + ChromaDB + OpenAI embeddings. Users upload PDF → ask questions.

Stack: LangChain, vector DBs, PDF parsing

AI Interview Prep Tool

Intermediate

Web app where students enter job role → AI generates 20 interview questions → grades answers → gives feedback. Builds with Streamlit + GPT-4 API. Actual product you can sell.

Stack: Python, OpenAI API, Streamlit, UI design

Multimodal AI App

Advanced

Build an app using GPT-4o or Gemini that accepts an image of a handwritten physics problem → solves it → explains step by step → generates similar practice problems.

Stack: Multimodal APIs, image processing, Python

AI Agent with Tools

Advanced

Build an agent using LangChain + OpenAI that can browse the web, read PDFs, write code, and execute it — to answer complex multi-step research questions autonomously.

Stack: LangChain agents, tool use, Python, APIs

Fine-tuned Specialist Model

Advanced

Fine-tune Llama 3 on domain-specific data (e.g., Indian legal judgements, medical literature, technical documentation). Deploy on Hugging Face. Open-source contribution.

Stack: PyTorch, transformers, LoRA fine-tuning, HuggingFace

Any Questions ???



By - Mohan KTK

Senior Trainer

You Are Now

GEN AI CERTIFIED!

Resources — Keep Learning:

→ chat.openai.com | claude.ai | gemini.google.com

→ deeplearning.ai | grow.google/ai-essentials

→ huggingface.co | kaggle.com | github.com

→ Newsletter: The Rundown AI | TLDR AI

→ YouTube: Andrej Karpathy | Two Minute Papers

#GenAIWorkshop | Build. Share. Grow.



By - Mohan KTK

Senior Trainer



CERTIFICATE OF COMPLETION

Awarded to

G vishnuvardhan

for successful completion of workshop on
GenAI Certified Workshop

from PrepInsta Prime
Issued on: **12-06-2026**

A handwritten signature in black ink, appearing to read 'Atulya Kaushik'.

Atulya Kaushik
Co-Founder, Director

This certificate is awarded to

GADIPE VISHNUVARDHAN

for successfully completing

Introduction to Cybersecurity

through the Cisco Networking Academy program.



Lynn Bloomer
Director
Cisco Networking Academy

25 Dec 2023
Completion Date

This certificate is awarded to

GADIPE VISHNUVARDHAN

for successfully completing

IT Essentials: PC Hardware and Software

through the Cisco Networking Academy program.



Lynn Bloomer
Director
Cisco Networking Academy

21 Mar 2022
Completion Date

COURSE CERTIFICATE

This is to certify that

G vishnuvardhan

has successfully completed the course by demonstrating
theoretical and practical understanding of

Introduction to Python



Yeva Hyusyan
Chief Executive Officer



Certificate CC-C1ZPZAAK
Issued 02 September, 2023

